

# Series

## Geometric Series (g-series)

$$\sum_{n=0}^{\infty} ar^n$$

• the "n" is a exponent this is important because it will help distinguish geometric series from the other series

• you can find the sum of the series if it diverges

$$Sum = \sum_{n=0}^{\infty} \frac{a}{1-r}$$

• this works only when  $n=0$  why, you have to remember for geometric series  $n$  is a exponent so  $ar^0 \Rightarrow ar^1 = a$

$$Sum = \sum_{n=1}^{\infty} \frac{ar^n}{1-r}$$

• this equation takes into consideration "n" can be equal to other values not just 0 so when  $n$  is equal to anything other than 0 use this equation you can also use this when  $n=0$

• a geometric series has a constant  $r$  value if the  $r$  value changes its **Not** a geometric series

• converges when  $-1 < r < 1$  when the  $r$ -value is between  $-1$  and  $1$

• diverges when  $r \leq -1$  or  $r \geq 1$  or you say that it diverges when when the  $r$ -value is not between  $-1$  and  $1$

$$0.46 + 0.0046 + 0.000046$$

$$r = \frac{\text{2nd term}}{\text{1st term}} = \frac{0.0046}{0.46} = \frac{1}{100}$$

this will get you the  $r$  value but what if we want to know if it's a geometric or not. We know that in a geometric series  $r$ -value is constant

$$r = \frac{\text{3rd term}}{\text{2nd term}} = \frac{0.000046}{0.0046} = \frac{1}{100}$$

• we checked using another set of term and we got the answer as we previously did we can say that is a geometric series

$$a = 0.46 \quad r = \frac{1}{100}$$

$-1 < r < 1$   
 $r$  is between  $-1$  and  $1$  it diverges

$$\frac{a}{1-r} = \frac{0.46}{1-\frac{1}{100}} = \boxed{\frac{46}{99}}$$

## Finding a-value

$$\sum_{n=0}^{\infty} \frac{3^n}{5^{n-1}} \Rightarrow \frac{3^2}{5^{2-1}} = \frac{9}{5}$$

• when you are dealing with problem like this plug the first number of the  $n$ -values

## P-series / harmonic - series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \star n \neq 0$$

• For P-series "n" will always be in the denominator this is important because it will help you distinguish P-Series from other series

• Converges if  $p > 1$

• diverges if  $p \leq 1$

$$\text{harmonic-series} \quad \sum_{n=1}^{\infty} \frac{1}{n}$$

• harmonic series occur when  $p=1$  all harmonic motion say is at  $p=1$  the series diverges that's it nothing else

there are no two conditional this occurs under one condition and only one condition that being  $p=1$

# integral test

When applying the integral test it must satisfy these condition these conditions must be satisfied before applying the test if they aren't met you cannot apply the integral test

- Positive
  - Continuous
  - decreasing
- these are not the test these are condition that must be met before applying the test

example

$$\int_1^{\infty} \frac{4}{9x-1}$$

Positive ✓

$x \geq 1$  • this will produce only positive  $x$  values

Continuous ✓

• to find if the equation you find the domain

$$9x-1 \neq 0$$

$$9x \neq 1$$

$$x \neq 1/9$$

•  $(-\infty, 1/9) \cup (1/9, \infty)$

the function is continuous

$1/9$  is not in  $x \geq 1$

decreasing ✓

• to find if the equation is decreasing you take the derivative if its negative its decreasing

$$\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}$$

$$f=4$$

$$\frac{df}{dx}=0$$

$$g=9x-1$$

$$\frac{dg}{dx}=9$$

$$\frac{(0)(9x-1) - (4)(9)}{(9x-1)^2}$$

$$= \frac{-36}{(9x-1)^2}$$

So it negative meaning decreasing

Sometimes you will be able to solve a series that looks like a integral test question this is completely fine the integral test should be your last resort if you can solve it with another method do so

## Divergence test

• this tests only for divergence not for convergence

$$\lim_{n \rightarrow \infty} a_n = 0 \text{ test fail}$$

$$\lim_{n \rightarrow \infty} a_n \neq 0 \text{ mean if you get an answer like } 2, 3, 5, 10 \text{ as long as its not zero it diverges}$$

Problem where you need to apply the divergence test don't happen often

# Comparison test

- if  $0 \leq a_n \leq b_n$  and  $\sum b_n$  Converges then  $\sum a_n$  Converge
- if  $0 \leq b_n \leq a_n$  and  $\sum b_n$  diverges then  $\sum a_n$  diverges
- its okay if some terms are negative as long as the series becomes positive and stays positive

## Case 1:

$$a_n = \frac{2^n - 7}{5^n}$$

- Negative at small  $n$  (like  $n = 1, n = 2$ )
- BUT for  $n \geq 3$ , it becomes positive and stays that way
- So: comparison test is OK if you start at  $n = 3$

## Case 2:

$$a_n = \frac{-1}{n}$$

- Always negative for every  $n \geq 1$
- So you can't use comparison test at all — it never becomes positive
- No comparison test allowed

you need to construct  $\sum b_n$  you can do that by selecting the leading term from the denominator and numerator of  $a_n$  once done you need to see if it converges or diverges you will most likely end up with geometric series or p-series depending on your result you can use the table to see the next step

if the test fails you must use limit comparison  
a comparison test looks like a p-series or geometric series just with another term added to it

## example

$$\sum_{n=1}^{\infty} \frac{2^n - 7}{5^n + 1} \quad a_n = \frac{2^n - 7}{5^n + 1} \quad b_n = \frac{2^n}{5^n} \quad \sum_{n=1}^{\infty} \left(\frac{2}{5}\right)^n$$

geometric series  
 $-1 < r = 0.4 < 1$   
Converges

Since  $b_n$  has converged we can look at the table above and so now we want to see if the following statement is true

$$a_n \leq b_n$$

$a_n$  less than or equal to  $b_n$  if true  $a_n$  converges

$$\frac{2^n - 7}{5^n + 1} \leq \frac{2^n}{5^n}$$

$$5^n(2^n - 7) \leq 2^n(5^n + 1)$$

$$5^n \cdot 2^n - 5^n \cdot 7 \leq 2^n \cdot 5^n + 2^n$$

$$-5^n \cdot 7 \leq 2^n$$

true

since all terms are positive in the inequality we can cross multiply. reconsider cross multiplication if they are negative

you need to write answer like this

$\sum b_n$  Converges and  $a_n \leq b_n$  for all " $n$ " values hence by comparison test  $\sum a_n$  Converges

# Limit Comparison test

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

if  $0 < L < \infty$

if  $\sum b_n$  Converges then  $\sum a_n$  Converges

if  $\sum b_n$  diverges then  $\sum a_n$  diverges

the limit comparison test requires positive sequences to work correctly

this test is usually done when the comparison test has failed but you can apply anytime you want but this can sometimes be tricky

example

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} \quad a_n = \frac{1}{n^2+1} \quad b_n = \frac{1}{n^2} \quad p=2 \text{ Converges because } p > 1$$

• p-series

if  $\sum b_n$  converges then  $\sum a_n$  converges

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^2+1} \cdot \frac{n^2}{1}$$

We know  $b_n$  converges and using limit comparison test  $L$  is between 0 and  $\infty$  so that mean  $\sum a_n$  converges both  $a_n$  and  $b_n$  converge

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = \frac{\infty}{\infty} \quad \text{apply l'hopital's rule}$$

$$\lim_{n \rightarrow \infty} \frac{2n}{2n} = 1 \approx 0 < \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1 < \infty$$

$L$  is between 0 and  $\infty$

## alternating Series

this is what an alternating series looks like

$$\sum (-1)^n \cdot b_n \text{ or } \sum \frac{(-1)^{n+1}}{2} \cdot b_n$$

$b_n$  is everything except  $(-1)^n$

Converges if

- $\lim_{n \rightarrow \infty} b_n = 0$

- $0 < b_{n+1} < b_n$  (or)  $b_n$  is decreasing (use derivative to show that it's decreasing)

Diverges by divergence test if

- $\lim_{n \rightarrow \infty} b_n \neq 0$

the reason we say that it diverge by divergence test because in other word we are kind of doing the divergence test

the way you can identify a alternating is alternating has  $(-1)^n$

example

$$\sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2+1} \quad b_n = \frac{4}{n^2+1}$$

$$\checkmark \lim_{n \rightarrow \infty} \frac{4}{n^2+1} = \frac{4}{\infty^2+1} = 0$$

$$\checkmark \frac{d}{dx} \left( \frac{4}{n^2+1} \right) = 4(-1)n^{-3} = -4(n^2+1)^{-2} \cdot 2n = \frac{-8n}{(n^2+1)^2} \text{ (decreasing)}$$

Converges by alternating Series

## this is usually done For alternating Series

$\sum a_n$  is absolutely Convergent IF  $\sum a_n$  Converges and  $\sum |a_n|$  Converges

$\sum a_n$  is Conditionally Convergent IF  $\sum a_n$  Converges and  $\sum |a_n|$  diverges

example

$$\sum \frac{(-1)^n}{n+1}$$

$$\sum \frac{a_n}{n+1} \quad b_n = \frac{1}{n+1}$$

$$\checkmark \lim_{n \rightarrow \infty} \frac{1}{n+1} = \frac{1}{\infty+1} = 0$$

$$\checkmark \frac{d}{dx} \left( \frac{1}{n+1} \right) \Rightarrow (n+1)^{-1} \Rightarrow -(n+1)^{-2} \approx -\frac{1}{(n+1)^2}$$

decreasing

Converges

$$\boxed{\sum \frac{(-1)^n}{n+1} = \text{Converges}}$$

$$\sum |a_n|$$

$$\sum \frac{1}{n+1}$$

$$a = \frac{1}{n+1}$$

$$b = \frac{1}{n}$$

• diverges by harmonic series

$$b_n \leq a_n$$

$$\frac{1}{n} \leq \frac{1}{n+1} = \frac{n+1}{n+1} \leq \frac{n}{n+1}$$

False

test fails so  
apply limit comparison

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{1}{n+1} \cdot \frac{n}{1} = \frac{n}{n+1} = \frac{\infty}{\infty} \quad \text{apply l'Hopital's rule}$$

$$\lim_{n \rightarrow \infty} \frac{1}{1} = 1 \quad 0 < \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1 < \infty$$

$$\sum a_n = \text{diverges}$$

$$\boxed{\sum \frac{1}{n+1} = \text{diverges}}$$

after Find  $\sum a_n$  and  $\sum |a_n|$  we Found  $a_n$  is Converging and  $\sum |a_n|$  is diverging  
Conditionally Converget

## Root test

the root test is the whole quantity raised to the power of  $n$ . the test works like this you place the  $|a_n|$  to the  $n$ -th root

- the root test can be identified by everything being raised to the power of  $n$

$$\sum_{n=1}^{\infty} (a+b)^n \Rightarrow |a+b|^n \Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{|a+b|^n}$$

- if answer less than 1 it converges
- if answer greater than 1 it diverges
- if answer equal to 1 the test fails (apply another test)

why the  $n$ -th root works

$$e^x: \sum_{n=1}^{\infty} \left(\frac{a}{a+b}\right)^n$$

$$\sqrt[n]{\left|\left(\frac{a}{a+b}\right)^n\right|} \approx \left(\frac{a}{a+b}\right)^{n/n} \Rightarrow \left(\frac{a}{a+b}\right)^1$$

## ratio test

$$\sum_{n=1}^{\infty} \frac{a^n}{n!}$$

the ratio test can be identified by having a " $n$ " in the power and one in the base

$$\approx \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\begin{aligned} n! &= n\text{-Factorial} \approx 1 \cdot 2 \cdot 3 \cdot (n-1) \cdot (n) \\ 0! &= 1 \\ (n+1)! &= n! \cdot (n+1) \end{aligned}$$

$(n+1)!$  is the same thing as  $n! \cdot (n+1)$  you cannot distribute the  $n!$  into  $(n+1)$  distribution does not apply to  $n$ -Factorial

For  $a_{n+1}$  you are only adding a one where ever there is a one and for  $a_n$  that will just be the original equation

$$1) \lim_{x \rightarrow \infty} (e^{5x} + x)^{1/x}$$

$$L = \lim_{x \rightarrow \infty} (e^{5x} + x)^{1/x}$$

$$\ln(L) = \lim_{x \rightarrow \infty} \frac{1}{x} \ln(e^{5x} + x)$$

$$\lim_{x \rightarrow \infty} \frac{\ln(e^{5x} + x)}{x} = \frac{\infty}{\infty} \text{ apply L'Hospital's rule}$$

$$\lim_{x \rightarrow \infty} \frac{5e^{5x} + 1}{e^{5x} + x} = \frac{\infty}{\infty} \text{ apply L'Hospital's rule again}$$

$$\lim_{x \rightarrow \infty} \frac{25e^{5x}}{5e^{5x} + 1} = \frac{\infty}{\infty} \text{ apply L'Hospital's rule again}$$

$$\lim_{x \rightarrow \infty} \frac{125e^{5x}}{25e^{5x}} = 5$$

We introduced a  $L$  we must bring it back

$$\ln(L) = 5$$

$$e^{\ln(L)} = e^5$$

$$\boxed{L = e^5}$$

$$3) \lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$$

$$L = \lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x \Rightarrow \ln(L) = \lim_{x \rightarrow \infty} x \ln\left(1 + \frac{3}{x}\right)$$

$$\lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{3}{x}\right)}{x^{-1}} = \frac{0}{0} \text{ apply L'Hospital's rule}$$

$$\lim_{x \rightarrow \infty} -\frac{3x^2}{1 + \frac{3}{x}} \cdot \frac{1}{x^2} = \lim_{x \rightarrow \infty} \frac{3}{1 + \frac{3}{x}} = 3$$

$$\ln(L) = 3$$

$$e^{\ln(L)} = e^3$$

$$\boxed{L = e^3}$$

$$2) \lim_{x \rightarrow 0^+} (\tan(2x))^x$$

$$L = \lim_{x \rightarrow 0^+} (\tan(2x))^x$$

$$\ln(L) = \lim_{x \rightarrow 0^+} x \cdot \ln(\tan(2x))$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(\tan(2x))}{x^{-1}} = \frac{0}{0} \text{ apply L'Hospital's rule}$$

$$\frac{2 \sec^2 x}{\tan^2 x} \Rightarrow \lim_{x \rightarrow 0^+} \frac{2x^2 \sec^2 x}{\tan^2(2x)} = \frac{0}{0} \text{ apply L'Hospital's rule}$$

$$f = 2x^2 \quad g = \sec^2(2x)$$

$$\frac{df}{dx} = 4x \quad \frac{dg}{dx} = 2 \sec(2x) \cdot \tan(2x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{(4x)(\sec^2(2x) + (2x^2)(2 \sec^2(2x) \tan(2x)))}{2 \sec^2(2x)} \Rightarrow \frac{\sec^2(2x)(4x + 2x^2 \tan(2x))}{\sec^2(2x)} = \frac{0}{1}$$

$$\ln(L) = 0$$

$$e^{\ln(L)} = e^0 \Rightarrow \boxed{L = e^0}$$

$$4) \int_2^3 \frac{dx}{(3-x)^{2/3}}$$

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ 2 \quad m \quad 3 \end{array}$$

$m$  approaches 3  
From the left  $m$  is a number in the middle and 3 is greater than  $m$

$$\lim_{m \rightarrow 3} \int_2^m \frac{1}{(3-x)^{2/3}} dx \quad \begin{array}{l} u = 3-x \\ du = -dx \\ -du = dx \end{array} \quad - \int_2^m \frac{1}{(u)^{2/3}} du$$

$$\lim_{m \rightarrow 3} -[3(3-x)^{1/3}]_2^m \Rightarrow -[3(3-m)^{1/3} - 3(3-2)^{1/3}]$$

$$\lim_{m \rightarrow 3} -3(3-3)^{1/3} + 3(1)^{1/3}$$

$$= \boxed{3}$$

converges to 3

$$5) \int_2^3 \frac{dx}{x^2-4}$$

m is approaching 2 from the right. m is a number in the middle and 2 is smaller than m

$$\lim_{m \rightarrow 2} \int_m^3 \frac{1}{x^2-4} dx \quad \text{• use Partial Fractions}$$

$$\int \frac{A}{(x-2)} + \int \frac{B}{(x+2)} \quad \begin{matrix} u = x-2 & u = x+2 \\ du = dx & du = dx \end{matrix}$$

$$A \int \frac{1}{u} + B \int \frac{1}{u} \Rightarrow A \ln|x-2| + B \ln|x+2|$$

$$1 = A(x+2) + B(x-2) \quad \text{• multiplying each term by LCD which is } (x+2)(x-2)$$

$$x @ 2 \Rightarrow 1 = A(2+2) + B(2-2) \quad \text{• x can be anything. You want to isolate and solve for A and B}$$

$$A = 1/4$$

$$x @ -2 \Rightarrow 1 = \frac{1}{4}(-2+2) + B(-2-2)$$

$$B = -1/4$$

$$\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

$$\frac{1}{4} \ln|x-2| - \frac{1}{4} \ln|x+2|$$

$$\lim_{m \rightarrow 2} \left[ \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| \right] \Rightarrow \lim_{m \rightarrow 2} \left[ \frac{1}{4} \ln \left| \frac{1}{5} \right| - \frac{1}{4} \ln \left| \frac{m-2}{m+2} \right| \right]$$

$$\frac{1}{4} \ln \left| \frac{1}{5} \right| - \ln \left| \frac{2-2}{2+2} \right| \Rightarrow \frac{1}{4} \ln \left| \frac{1}{5} \right| - (-\infty) = \text{diverges}$$

$$6) \sum_{n=2}^{\infty} \frac{3^n}{4^{n+1}} \approx \sum_{n=2}^{\infty} \frac{3^n}{4^n \cdot 4}$$

$$\sum_{n=2}^{\infty} \frac{1}{4} \left( \frac{3}{4} \right)^n \quad \begin{matrix} \text{• geometric series} \\ r = 0.75 \\ -1 < 0.75 < 1 \\ \text{converges} \end{matrix}$$

$$\frac{ar^n}{1-r} = \frac{1/4 \cdot (3/4)^2}{1-(3/4)} = \frac{9/64}{1/4} = \frac{9}{16}$$

$$6) \int_2^{\infty} \frac{1}{x e^{\sqrt{x}}} dx$$

$$u = \sqrt{x}$$

$$\lim_{m \rightarrow \infty} \int_2^m \frac{1}{x e^{\sqrt{x}}} dx$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$2\sqrt{x} du = dx$$

$$\lim_{m \rightarrow \infty} \int_2^m \frac{1}{x e^u} \cdot 2\sqrt{x} du$$

$$\lim_{m \rightarrow \infty} 2 \int_2^m \frac{1}{e^u} du \Rightarrow \lim_{m \rightarrow \infty} \left[ \frac{2}{e^u} \right]_2^m$$

$$\lim_{m \rightarrow \infty} \left[ \frac{2}{e^m} - \frac{2}{e^2} \right]$$

$$\lim_{m \rightarrow \infty} \frac{2}{e^m} - \frac{2}{e^2} = \frac{2}{e^2}$$

$$7) \int_{-\infty}^{-5} \frac{1}{x^2+x} dx$$

$$\lim_{m \rightarrow -\infty} \int_m^{-5} \frac{1}{x(x+1)} dx$$

$$u = x+1$$

$$du = dx$$

$$\int \frac{A}{x} + \int \frac{B}{(x+1)} \Rightarrow A \ln|x| + B \ln|u|$$

$$A \ln|x| + B \ln|x+1|$$

$$\left[ \ln|x| - \ln|x+1| \right]_m^{-5}$$

$$1 = A(x+1) + Bx$$

$$x @ 0 \Rightarrow 1 = A(0+1) + B(0)$$

$$A = 1$$

$$x @ -1 \Rightarrow 1 = 1(-1+1) + B(-1)$$

$$B = -1$$

$$\lim_{m \rightarrow -\infty} \left[ \ln \left| \frac{x}{x+1} \right| \right]_m^{-5} \Rightarrow \lim_{m \rightarrow -\infty} \left[ \ln \left| \frac{-5}{-4} \right| - \ln \left| \frac{m}{m+1} \right| \right]$$

$$= \ln \left| \frac{5}{4} \right|$$

there is no need to apply l'Hopital's rule it will eventually go to  $\ln(1) = 0$



$$9) \sum_{n=2}^{\infty} \frac{5-2^n}{3^{n-2}} \approx \sum_{n=2}^{\infty} \frac{5}{3^{n-2}} - \sum_{n=2}^{\infty} \frac{2^n}{3^{n-2}}$$

$$\sum_{n=2}^{\infty} \frac{5}{3^{n-2}} - \sum_{n=2}^{\infty} \frac{2^n}{3^{n-2}}$$

$$\sum_{n=2}^{\infty} \frac{5}{3^{n-2}} - \sum_{n=2}^{\infty} \frac{2^n}{3^{n-2}}$$

$$\sum_{n=2}^{\infty} 45 \cdot \left(\frac{1}{3}\right)^n - \sum_{n=2}^{\infty} 4 \left(\frac{2}{3}\right)^n$$

• geometric  
r = 1/3  
-1 < r = 1/3 < 1  
Converges

$$\frac{ar^n}{1-r} = \frac{45(1/3)^2}{1-(1/3)} = \frac{5}{2}$$

$$\frac{15}{2} - 12 = \frac{-9}{2}$$

• geometric

$$r = 2/3$$

$$-1 < 2/3 < 1$$

converges

$$\frac{ar^n}{1-r} = \frac{4(2/3)^2}{1-2/3} = \frac{4}{1/3} = 12$$

• When you have a situation like this they both must either converge or diverge

$$10) \sum_{n=1}^{\infty} \frac{(-1)^n}{n 4^n}$$

• alternating Series

$$b_n = \frac{1}{n 4^n}$$

$$\lim_{n \rightarrow \infty} b_n \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n 4^n} = \frac{1}{\infty 4^n} = 0$$

$$\frac{d}{dx}(b_n) \Rightarrow \frac{d}{dx} \left( \frac{1}{n 4^n} \right) \Rightarrow (n \cdot 4^n)^{-1} \approx -1 (n \cdot 4^n)^{-2}$$

$$\frac{df}{dn} \cdot g + f \cdot \frac{dg}{dn} \quad f=n \quad g=4^n \quad \frac{df}{dn}=1 \quad \frac{dg}{dn}=4^n \cdot \ln(4)$$

$$(1) \cdot (4^n) + (n)(4^n \cdot \ln(4)) - (n \cdot 4^n)^2 \cdot 4^n + 4^n \cdot n \cdot \ln(4) = \frac{-(4^n + 4^n n \cdot \ln(4))}{(n \cdot 4^n)^2}$$

bn goes to 0 when we take lim n to infinity and is decreasing this makes it converging by the alternating series test

$$11) \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

$$b_n = \frac{1}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} b_n \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}} = 0$$

$$\frac{d}{dx} \left( \frac{1}{\sqrt{n+1}} \right) \Rightarrow (n+1)^{-1/2} \approx -\frac{1}{2} \cdot (n+1)^{-3/2} = \frac{-1}{2(n+1)^{3/2}}$$

• bn goes to 0 when we take n to infinity and is decreasing (negative) when we take the derivative this means that it converging

$$13) \sum_{n=1}^{\infty} \frac{7^n}{5^{n+1}} \quad a_n = \frac{7^n}{5^{n+1}}$$

$$b_n = \frac{7^n}{5^n} = \sum_{n=1}^{\infty} \left(\frac{7}{5}\right)^n$$

• geometric series

$$r = 1.4$$

diverges because its not in between -1 and 1

apply limit comparison

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{7^n}{5^{n+1}} \cdot \frac{5^n}{7^n} = \frac{\infty}{\infty} \quad \text{apply L'Hospital rule}$$

$$\text{the } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1 \text{ meaning}$$

it is between 0 and infinity

the series become diverging

$$\lim_{n \rightarrow \infty} \frac{35^n / \ln(35)}{35^n / \ln(35)} = 1$$

$$12) \sum_{n=2}^{\infty} \frac{\sqrt[3]{n}}{\sqrt{n}} \approx \sum_{n=2}^{\infty} \frac{n^{1/3}}{n^{1/2}} = n^{1/3} \cdot n^{-1/2}$$

• p-Series

$$p = 1/6 < 1 \text{ (less than one)}$$

$$\sum_{n=2}^{\infty} \frac{1}{n^{1/6}}$$

diverging by p-series test

$$14) \sum_{n=1}^{\infty} \frac{7^n}{5^{n+1}}$$

$$a_n = \frac{7^n}{5^{n+1}}$$

$$b_n = \frac{7^n}{5^n} = \sum_{n=1}^{\infty} \left(\frac{7}{5}\right)^n$$

$$b_n \leq a_n \quad \text{• we did this because } b_n \text{ diverges}$$

$$\frac{7^n}{5^n} \leq \frac{7^n}{5^{n+1}} \approx 3 \cdot 5^n + 7^n \leq 3 \cdot 5^n$$

$$7^n \leq 0$$

False!!

apply limit Comparison

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \Rightarrow \lim_{n \rightarrow \infty} \frac{7^n}{5^{n+1}} \cdot \frac{5^n}{7^n} \Rightarrow \lim_{n \rightarrow \infty} \frac{5^n}{5^{n+1}} = \frac{\infty}{\infty} \quad \text{apply L'Hospital's rule}$$

$$\text{LH\#} \lim_{n \rightarrow \infty} \frac{\ln(5^n) \cdot 5^n}{\ln(5^n) \cdot 5^n} = 1$$

$$0 < \lim_{n \rightarrow \infty} \frac{a_n}{b_n} < \infty$$

• a\_n diverges because it follows the b\_n statement  
What ever b\_n is a\_n will be the same as long as the limit comparison is between 0 and ∞

the series diverges

$$15) \sum_{n=1}^{\infty} n \cdot e^{-n} \approx \sum_{n=1}^{\infty} \frac{n}{e^n}$$

$$a_n = \frac{n}{e^n}$$

$$a_{n+1} = \frac{(n+1)}{e^{n+1}} \approx \frac{(n+1)}{e^n \cdot e}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \approx \lim_{n \rightarrow \infty} \frac{(n+1)}{e^n \cdot e} \cdot \frac{e^n}{n} \approx \frac{(n+1)}{e \cdot n}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)}{e n} = \frac{\infty}{\infty} \quad \text{• apply L'Hospital's rule}$$

$$\text{LH\#} \lim_{n \rightarrow \infty} \frac{1}{e} = \frac{1}{e}$$

•  $0 \leq r < 1$  abs converges  
•  $r = 1$  test fails  
•  $r > 1$  diverges

$$\frac{1}{e} = 0.367 \dots$$

the series absolutely converges

$$16) \sum_{n=1}^{\infty} \frac{2n}{n^3-1}$$

$$a_n = \frac{2n}{n^3-1}$$

$$b_n = \frac{2n}{n^3} \Rightarrow \sum_{n=1}^{\infty} \frac{2}{n^2} \quad \text{• p-series } p=2 > 1 \text{ so } b_n \text{ converges}$$

We apply the limit comparison because it will be easier then continuing but keep in mind you can apply the limit comparison whenever you want you don't have to use it when the comparison test fails

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \Rightarrow \lim_{n \rightarrow \infty} \frac{2n}{n^3-1} \cdot \frac{n^3}{2n} \approx \frac{n^3}{n^3-1}$$

$$\lim_{n \rightarrow \infty} \frac{n^3}{n^3-1} = \frac{\infty}{\infty} \quad \text{apply L'Hospital's rule}$$

$$0 < \lim_{n \rightarrow \infty} \frac{a_n}{b_n} < \infty$$

$$\text{LH\#} \lim_{n \rightarrow \infty} \frac{3n^2}{3n^2} = 1$$

a\_n converge like b\_n

the series converges

$$17) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+1}$$

$$b_n = \frac{1}{n^2+1}$$

$$\lim_{n \rightarrow \infty} b_n \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^2+1} = 0$$

$$\Rightarrow \frac{d}{dn} (n^2+1)^{-1} = -\frac{2n}{(n^2+1)^2}$$

• both condition  $\lim_{n \rightarrow \infty} b_n = 0$  and decreasing

the series absolutely converges by the alternating series test

$$18) \sum_{n=1}^{\infty} \frac{3^n}{n!}$$

$$a_n = \frac{3^n}{n!}$$

$$a_{n+1} = \frac{3^{n+1}}{n!(n+1)} \Rightarrow \frac{3^n \cdot 3^1}{n! \cdot (n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{3^n \cdot 3^1}{n! \cdot (n+1)} \cdot \frac{n!}{3^n} \right|$$

$$\lim_{n \rightarrow \infty} \frac{3^1}{n+1} = 0$$

- $0 \leq r < 1$  absolutely converging
- $r = 1$  test fails
- $r > 1$  diverges

the series absolutely converges by the ratio test

$$19) \sum_{n=1}^{\infty} \left( \frac{3+2n}{n+1} \right)^n$$

$$\sqrt[n]{\left| \left( \frac{3+2n}{n+1} \right)^n \right|}$$

$$\text{LH \#1} \quad \lim_{n \rightarrow \infty} \frac{2}{1} = 2$$

$$\approx \lim_{n \rightarrow \infty} \frac{3+2n}{n+1} = \frac{\infty}{\infty} \quad \text{apply L'Hopital's rule}$$

- $0 \leq r < 1$  absolutely converges
- $r = 1$  test fails
- $r > 1$  diverges

the test diverges by the root test

$$20) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}}$$

$$a_n = \frac{1}{\sqrt{n+1}}$$

$$b_n = \frac{1}{\sqrt{n}} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

- P-Series
- $p = 0.5 < 1$
- $b_n$  diverges by P-series test

$$b_n \leq a_n$$

$$\frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{n+1}} \Rightarrow \sqrt{n+1} \leq \sqrt{n}$$

• apply limit comparison test

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{1}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{1} \Rightarrow \frac{\sqrt{n}}{\sqrt{n+1}} = \frac{\infty}{\infty} \quad \text{• apply L'Hopital's rule}$$

$$\text{LH \#1} \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\frac{1}{2} n^{-1/2}}{\frac{1}{2} (n+1)^{-1/2}} \Rightarrow \frac{(n+1)^{1/2}}{(n)^{1/2}}$$

$$\lim_{n \rightarrow \infty} \left( \frac{n+1}{n} \right)^{1/2} \Rightarrow \left( \frac{n}{n} + \frac{1}{n} \right)^{1/2} \Rightarrow \left( 1 + \frac{1}{n} \right)^{1/2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^{1/2} \Rightarrow (1)^{1/2} = 1$$

$$0 < \lim_{n \rightarrow \infty} \frac{a_n}{b_n} < \infty$$

$b_n$  diverges so does  $a_n$ . So the whole thing diverges

$$21) \sum_{n=1}^{\infty} (2/3) \Rightarrow \sum_{n=1}^{\infty} \left( \frac{2}{3} \right) (1)^n$$

- diverges because  $r = 1$
- $r$  has to be  $-1 < r < 1$  so this does not work

Series diverges by the geometric series test

- this will keep on going on so there really no point of doing L'Hopital's rule again you need to simplify it more